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Program:

```
1  -- Step 1: Create Physician Table
2  ● CREATE DATABASE physician_department_db;
3  ● USE physician_department_db;
4  ● CREATE TABLE physician (
5      employeeid INT PRIMARY KEY,
6      name VARCHAR(100),
7      position VARCHAR(100),
8      ssn VARCHAR(20)
9  );
10 -- Step 2: Insert Data into Physician
11 ● INSERT INTO physician VALUES
12 (1,'Tejaswini','Staff Internist','11111111'),
13 (2,'Ravinder','Attending Physician','22222222'),
14 (3,'Srilatha','Surgical Attending Physician','33333333'),
15 (4,'Santhvin','Senior Attending Physician','44444444'),
16 (5,'Varshika','Head Chief of Medicine','55555555'),
17 (6,'Sahasra','Surgical Attending Physician','66666666'),
18 (7,'Ravi','Surgical Attending Physician','77777777'),
19 (8,'Pranitha','MD Resident','88888888'),
20 (9,'Angela','Attending Psychiatrist','99999999');
```

Output:

#	Time	Action	Message	Duration / Fetch
✓ 1	18:48:12	CREATE DATABASE physician_department_db	1 row(s) affected	0.032 sec
✓ 2	18:48:15	USE physician_department_db	0 row(s) affected	0.000 sec
✓ 3	18:48:17	CREATE TABLE physician (employeeid INT PRIMARY KEY, na...	0 row(s) affected	0.047 sec
✓ 4	18:48:20	INSERT INTO physician VALUES (1,'Tejaswini','Staff Internist','11111...	9 row(s) affected Records: 9 Duplicates: 0 Warnings: 0	0.016 sec

Program:

```
21 -- Step 3: Create Department Table
22 • CREATE TABLE department (
23     departmentid INT PRIMARY KEY,
24     name VARCHAR(100),
25     head INT
26 );
27 -- Step 4: Insert Data into Department
28 • INSERT INTO department VALUES
29     (1,'General Medicine',4),
30     (2,'Surgery',7),
31     (3,'Psychiatry',9);
```

Output:

✓	5	18:49:58	CREATE TABLE department (departmentid INT PRIMARY KEY, ...	0 row(s) affected	0.063 sec
✓	6	18:50:01	INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surg...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec

Program:

```
32 -- Step 5: INNER JOIN
33 • SELECT d.name AS Department,
34     p.name AS Head_Physician
35 FROM department d
36 JOIN physician p
37 ON d.head = p.employeeid;
```

Output:

Result Grid		Filter Rows:
	Department	Head_Physician
▶	General Medicine	Santhvin
	Surgery	Ravi
	Psychiatry	Angela

✓	7	18:51:05	SELECT d.name AS Department, p.name AS Head_Physician ...	3 row(s) returned	0.000 sec / 0.000 sec
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Program:

```
38      -- Step 6: WHERE Clause
39  ●    SELECT *
40      FROM physician
41      WHERE position = 'Surgical Attending Physician';
```

Output:

Result Grid					Filter Rows:	Edit:
	employeeid	name	position	ssn		
▶	3	Srilatha	Surgical Attending Physician	333333333		
	6	Sahasra	Surgical Attending Physician	666666666		
	7	Ravi	Surgical Attending Physician	777777777		
•	NULL	NULL	NULL	NULL		

8 18:52:30 SELECT * FROM physician WHERE position = 'Surgical Attending ... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
42      -- Step 7: WHERE + LIKE
43  ●    SELECT *
44      FROM physician
45      WHERE name LIKE 'John%';
```

Output:

Result Grid					Filter Rows:
	employeeid	name	position	ssn	
•	NULL	NULL	NULL	NULL	

9 18:53:24 SELECT * FROM physician WHERE name LIKE 'John%' LIMIT 0, ... 0 row(s) returned 0.000 sec / 0.000 sec

Program:

```
46      -- Step 8: COUNT()
47  ●    SELECT COUNT(*) AS Total_Physicians
48      FROM physician;
```

Output:

Result Grid		Filter Rows:
	Total_Physicians	
▶	9	

✓ 10 18:54:22 SELECT COUNT(*) AS Total_Physicians FROM physician LIMIT 0,... 1 row(s) returned 0.016 sec / 0.000 sec

Program:

```
49 -- Step 9: COUNT DISTINCT
50 • SELECT COUNT(DISTINCT position) AS Different_Positions
51 FROM physician;
```

Output:

Result Grid		Filter
	Different_Positions	
▶	7	

✓ 11 18:55:11 SELECT COUNT(DISTINCT position) AS Different_Positions FRO... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
52 -- Step 10: GROUP BY
53 • SELECT position,
54         COUNT(*) AS Total_Employees
55 FROM physician
56 GROUP BY position;
```

Output:

Result Grid		Filter Rows:
	position	Total_Employees
▶	Staff Internist	1
	Attending Physician	1
	Surgical Attending Physician	3
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

✓ 12 18:55:59 SELECT position, COUNT(*) AS Total_Employees FROM phys... 7 row(s) returned 0.000 sec / 0.000 sec

Program:

```
57      -- Step 11: GROUP BY + HAVING
58  •    SELECT position,
59          COUNT(*) AS Total_Employees
60  FROM physician
61  GROUP BY position
62  HAVING COUNT(*) > 1;
```

Output:

Result Grid




Filter Rows:

	position	Total_Employees
▶	Surgical Attending Physician	3



13 18:56:55

SELECT position, COUNT(*) AS Total_Employees FROM phys...

1 row(s) returned

0.000 sec / 0.000 sec

Program:

```
63      -- Step 12: ORDER BY ASC
64  •    SELECT *
65  FROM physician
66  ORDER BY name ASC;
```

Output:

Result Grid		Filter Rows:	Edit:
employeeid	name	position	ssn
9	Angela	Attending Psychiatrist	999999999
8	Pranitha	MD Resident	888888888
7	Ravi	Surgical Attending Physician	777777777
2	Ravinder	Attending Physician	222222222
6	Sahasra	Surgical Attending Physician	666666666
4	Santhvin	Senior Attending Physician	444444444
3	Srilatha	Surgical Attending Physician	333333333
1	Tejaswini	Staff Internist	111111111
5	Varshika	Head Chief of Medicine	555555555
NULL	NULL	NULL	NULL

14 18:57:52 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec

Program:

```
67      -- Step 13: ORDER BY DESC
68 •    SELECT *
69      FROM physician
70      ORDER BY employeeid DESC;
```

Output:

Result Grid

Filter Rows:

Edit:

	employeeid	name	position	ssn
▶	9	Angela	Attending Psychiatrist	999999999
	8	Pranitha	MD Resident	888888888
	7	Ravi	Surgical Attending Physician	777777777
	6	Sahasra	Surgical Attending Physician	666666666
	5	Varshika	Head Chief of Medicine	555555555
	4	Santhvin	Senior Attending Physician	444444444
	3	Srilatha	Surgical Attending Physician	333333333
	2	Ravinder	Attending Physician	222222222
	1	Tejaswini	Staff Internist	111111111
✱	NULL	NULL	NULL	NULL

15 18:58:56 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0... 9 row(s) returned

Program:

```
71      -- Step 14: WHERE + IN
72 •    SELECT *
73      FROM physician
74      WHERE employeeid IN
75      (
76          SELECT head
77          FROM department
78      );
```

Output:

	employeeid	name	position	ssn
▶	4	Santhvin	Senior Attending Physician	444444444
	7	Ravi	Surgical Attending Physician	777777777
	9	Angela	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

16 18:59:48 SELECT * FROM physician WHERE employeeid IN (SELECT h... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
79  -- Step 15: NOT IN
80  •  SELECT *
81  FROM physician
82  WHERE employeeid NOT IN
83  (
84      SELECT head
85      FROM department
86  );
```

Output:

	employeeid	name	position	ssn
▶	1	Tejaswini	Staff Internist	111111111
	2	Ravinder	Attending Physician	222222222
	3	Srilatha	Surgical Attending Physician	333333333
	5	Varshika	Head Chief of Medicine	555555555
	6	Sahasra	Surgical Attending Physician	666666666
	8	Pranitha	MD Resident	888888888
*	NULL	NULL	NULL	NULL

17 19:00:54 SELECT * FROM physician WHERE employeeid NOT IN (SEL... 6 row(s) returned 0.000 sec / 0.000 sec

Program:

```
87      -- Step 16: GROUP BY + ORDER BY
88  •    SELECT position,
89          COUNT(*) AS Total_Employees
90  FROM physician
91  GROUP BY position
92  ORDER BY Total_Employees DESC;
```

Output:

	position	Total_Employees
▶	Surgical Attending Physician	3
	Staff Internist	1
	Attending Physician	1
	Senior Attending Physician	1
	Head Chief of Medicine	1
	MD Resident	1
	Attending Psychiatrist	1

18 19:02:00 SELECT position, COUNT(*) AS Total_Employees FROM phys... 7 row(s) returned

0.000 sec / 0.000 sec

Program:

```
93      -- Step 17: WHERE + LIKE
94  •    SELECT *
95  FROM physician
96  WHERE position LIKE '%Attending%';
```

Output:

	employeeid	name	position	ssn
▶	2	Ravinder	Attending Physician	222222222
	3	Srilatha	Surgical Attending Physician	333333333
	4	Santhvin	Senior Attending Physician	444444444
	6	Sahasra	Surgical Attending Physician	666666666
	7	Ravi	Surgical Attending Physician	777777777
	9	Angela	Attending Psychiatrist	999999999
✱	NULL	NULL	NULL	NULL

19 19:03:36 SELECT * FROM physician WHERE position LIKE "%Attending%" L... 6 row(s) returned

0.000 sec / 0.000 sec

Program:

```
97      -- Step 18: JOIN + WHERE
98  •    SELECT p.name
99      FROM physician p
100     JOIN department d
101     ON p.employeeid = d.head
102     WHERE d.name = 'Surgery';
```

Output:

Result Grid	
	name
▶	Ravi

20 19:04:32 SELECT p.name FROM physician p JOIN department d ON p.empl... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
103      -- Step 19: JOIN + ORDER BY
104  •    SELECT d.name AS Department,
105          p.name AS Head_Physician
106     FROM department d
107     JOIN physician p
108     ON d.head = p.employeeid
109     ORDER BY d.name;
```

Output:

Result Grid		Filter Rows:
	Department	Head_Physician
▶	General Medicine	Santhvin
	Psychiatry	Angela
	Surgery	Ravi

21 19:05:23 SELECT d.name AS Department, p.name AS Head_Physician ... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
110      -- Step 20: WHERE + NOT LIKE
111  •    SELECT *
112     FROM physician
113     WHERE position NOT LIKE '%Surgical%';
```

Output:

	employeeid	name	position	ssn
▶	1	Tejaswini	Staff Internist	111111111
	2	Ravinder	Attending Physician	222222222
	4	Santhvin	Senior Attending Physician	444444444
	5	Varshika	Head Chief of Medicine	555555555
	8	Pranitha	MD Resident	888888888
	9	Angela	Attending Psychiatrist	999999999
*	NULL	NULL	NULL	NULL

22 19:06:18 SELECT * FROM physician WHERE position NOT LIKE '%Surgical...' 6 row(s) returned 0.000 sec / 0.000 sec

Program:

```
114 -- Step 21: COUNT + HAVING
115 • SELECT position,
116       COUNT(*) AS Total
117 FROM physician
118 GROUP BY position
119 HAVING COUNT(*) >= 2;
```

Output:

Result Grid	Filter Rows:
position	Total
▶ Surgical Attending Physician	3

23 19:07:26 SELECT position, COUNT(*) AS Total FROM physician GROU... 1 row(s) returned 0.000 sec / 0.000 sec

Program:

```
120 -- Step 22: EXISTS
121 • SELECT *
122 FROM department d
123 WHERE EXISTS
124 (
125     SELECT 1
126     FROM physician p
127     WHERE p.employeeid = d.head
128 );
```

Output:

	departmentid	name	head
▶	1	General Medicine	4
	2	Surgery	7
	3	Psychiatry	9
•	NULL	NULL	NULL

24 19:08:40 SELECT * FROM department d WHERE EXISTS (SELECT 1 ... 3 row(s) returned 0.000 sec / 0.000 sec

Program:

```
129      -- Step 23: MAX()
130 •    SELECT MAX(employeeid) AS Highest_Employee_ID
131      FROM physician;
```

Output:

Result Grid		Filter R
	Highest_Employee_ID	
▶	9	

25 19:09:31 SELECT MAX(employeeid) AS Highest_Employee_ID FROM physi... 1 row(s) returned 0.015 sec / 0.000 sec

Program:

```
132      -- Step 24: MIN()
133 •    SELECT MIN(employeeid) AS Lowest_Employee_ID
134      FROM physician;
```

Output:

Result Grid			Filter Row
	Lowest_Employee_ID		
	1		

26 19:10:23 SELECT MIN(employeeid) AS Lowest_Employee_ID FROM physi... 1 row(s) returned 0.000 sec / 0.000 sec